DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

LABORATORY OBSERVATION RECORD

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Student Name:	Maddala Surya Prabhath
Roll No:	A24126552094
Branch & Sec:	CSM - B
Date of Execution:	01-08-2026
Course Name:	Full Stack Web Development
Course Code:	23CM4121

Task: 01

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1. TITLE OF THE EXPERIMENT:

College Information Portal: Implementation of Fundamental HTML5 Structure, Tables, Forms, and Semantic Multimedia Features

2. MAPPED OUTCOMES:

• Course Outcome (CO): CO1 - Develop well-structured, modern HTML5 web pages using tables, forms, multimedia, and semantic elements.

• Program Outcomes (POs): PO1 (Engineering Knowledge), PO3 (Design/Development of Solutions), PO5 (Modern Tool Usage).

3. AIM:

To design and construct a comprehensive 'College Information Portal' webpage using HTML5, incorporating standard document hierarchy, text formatting, hyperlinks, structured tables, interactive registration forms, semantic layout tags, multimedia elements (<audio>, <video>), and specialized HTML5 input controls.

4. ALGORITHM / DESIGN:

- Step 1: Define a valid HTML5 template starting with <!DOCTYPE html> and basic structure (<html>, <head>, <body>).
- Step 2: Structure the layout using HTML5 semantic elements (<header>, <nav>, <section>, <article>, <footer>).
- Step 3: Add headings, formatted text (<b>, <i>, <u>, <sup>, <sub>), hyperlinks (to college site & Google), and ordered/unordered lists.
- Step 4: Build a Student Details table containing columns for Roll No, Name, Branch, and Email using <table>, <tr>, <th>, and <td> tags.
- Step 5: Insert college logo and content images with descriptive alt attributes.
- Step 6: Construct a Student Registration Form featuring standard inputs (Name, Password), radio buttons (Gender), checkboxes (Hobbies), dropdown (<select> for Branch), <textarea> for Comments, and HTML5 control types (Email, Date, Number, Color, Range).
- Step 7: Embed native <audio> and <video> tags with control attributes and verify layout presentation across browsers.

5. TEST CASES & EXECUTION DATA:

TC ID	Input / Module Test	Expected Result	Status
TC-01	HTML5 Semantic Structure & Text Elements	Header, Nav, Section, Lists & formatted text render cleanly without syntax bugs.	**Passed**
TC-02	Student Details Table & Images	Table displays Roll No/Name/Branch/Email correctly; Images load with proper alt text.	**Passed**
TC-03	Student Registration Form & HTML5 Controls	Radio buttons, checkboxes, dropdown, textarea, and HTML5 pickers (Date, Color, Range) capture user input.	**Passed**
TC-04	Multimedia Embedding (<audio> & <video>)	Audio and video players initialize with play/pause controls	**Passed**

functional.

6. OBSERVATION INTERVIEW QUESTIONS & ANSWERS:

Q1: What is the purpose of semantic HTML5 elements?

Ans: Semantic HTML5 elements (<header>, <nav>, <section>, <article>, <footer>) explicitly describe their meaning to both the browser and developer. They improve page accessibility, SEO optimization, document structure clarity, and code maintainability compared to generic <div> tags.

Q2: Compare Ordered and Unordered Lists.

Ans: An Ordered List (<ol>) displays items in a sequential numbered/alphabetical order where item sequence matters. An Unordered List (<ul>) renders items with bullet points where sequence is arbitrary.

Q3: What happens if an image path is incorrect?

Ans: If the src path of an <img> tag is invalid or broken, the browser displays a broken image icon alongside the text specified in the alt (alternate text) attribute.

Q4: Differentiate Radio Buttons and Checkboxes.

Ans: Radio Buttons (type="radio") allow the user to select only ONE option from a mutually exclusive group. Checkboxes (type="checkbox") allow selecting MULTIPLE independent options simultaneously.

Q5: Explain the advantages of HTML5 input types over traditional text boxes.

Ans: HTML5 input types (email, date, number, color, range) provide built-in client-side validation, open native UI pickers (date/color sliders), and trigger optimized mobile keyboards automatically without requiring JavaScript code.

7. OUTPUT PROOF:

Repository Code & Task Files Link:

https://github.com/maddalasuryaprabhath/fs_lab/tree/main/Observation/observation_task_1

8. LEARNING OUTCOME / RESULT:

Successfully developed and verified the 'College Information Portal' webpage. Understood and demonstrated competence in HTML5 document structuring, tabular data presentation, form control design, native multimedia embedding, and semantic layout architecture.

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LABORATORY OBSERVATION RECORD

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Student Name:	Maddala Surya Prabhath
Roll No:	A24126552094
Branch & Sec:	CSM - B
Date of Execution:	01-08-2026
Course Name:	Full Stack Web Development / Java Programming
Course Code:	23CM4121

Task: 02

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1. TITLE OF THE EXPERIMENT:

My Styled Webpage: Dynamic HTML5 Canvas Graphics Rendering and CSS3 Web Layout Styling

2. MAPPED OUTCOMES:

• Course Outcome (CO): CO1 - Draw basic graphics using HTML5 Canvas, style web pages using CSS3, construct visually appealing card layouts, and apply modern hover effects.

• Program Outcomes (POs): PO1 (Engineering Knowledge), PO3 (Design/Development of Solutions), PO5 (Modern Tool Usage).

3. AIM:

To design and construct 'My Styled Webpage' incorporating programmatic graphics drawing on an HTML5 <canvas> element (rectangle, circle, line, and custom text) and applying CSS3 styling rules (gradients, box/text shadows, border-radius, margins, padding, and pseudo-class hover effects) across a structured card layout.

4. ALGORITHM / DESIGN:

- Step 1: Create an HTML5 document structure containing <header>, <nav>, main layout section with three <article> content cards, <canvas> container, and <footer>.
- Step 2: Initialize JavaScript context (getContext('2d')) for the <canvas> element.
- Step 3: Programmatically render shapes on Canvas: draw a filled/outlined rectangle (fillRect), a circle (arc), a line segment (lineTo), and display text (fillText) for student name.
- Step 4: Formulate CSS3 stylesheet applying gradient background (linear-gradient), custom font families, font sizes, and text alignments.
- Step 5: Apply Box Model styling properties including margins, paddings, border-radius, border, box-shadow, and text-shadow to content cards and headers.
- Step 6: Implement interactive dynamic visual feedback using CSS pseudo-classes (:hover) on navigation links and action buttons.
- Step 7: Verify responsive graphics rendering and dynamic visual interactivity in the web browser.

5. TEST CASES & EXECUTION DATA:

TC ID	Input / Module Test	Expected Result	Status
TC-01	HTML5 Canvas Graphics Rendering	Rectangle, circle, line segment, and student name text render accurately on canvas.	Passed
TC-02	CSS3 Layout & Styling Properties	Gradient background, font typography, card borders, rounded corners, and shadows apply correctly.	Passed
TC-03	Interactive Pseudo-Class Hover Effects	Navigation links and buttons smoothly transition color, scale, or shadow on mouse hover.	Passed

6. OBSERVATION INTERVIEW QUESTIONS & ANSWERS:

Q1: Differentiate Margin and Padding.

Ans: Margin is the space outside an element's border that separates it from adjacent elements. Padding is the space inside an element's border between the border line and the internal content.

Q2: Explain the difference between Border and Outline.

Ans: A Border is part of the element's box model and takes up space, affecting element dimensions. An Outline is drawn outside the border, does NOT take up layout space, and does not alter element dimensions.

Q3: What is the purpose of the Canvas element?

Ans: The <canvas> element provides a resolution-dependent bitmap canvas area where graphics, shapes, animations, and game visual components can be drawn dynamically using JavaScript APIs.

Q4: Compare Inline, Internal, and External CSS.

Ans: Inline CSS uses the style attribute directly on HTML elements (highest priority, hard to maintain). Internal CSS uses <style> tags within the <head> of a single document. External CSS uses an independent .css file linked via <link rel="stylesheet"> (best practice for reusability).

Q5: How do Hover effects improve user experience?

Ans: Hover effects provide immediate visual feedback (affordance) to users, indicating that an element (like a button or link) is clickable and interactive, which improves navigational clarity and UX engagement.

7. OUTPUT PROOF:

Repository Code & Task Files Link:

https://github.com/maddalasuryaprabhath/fs_lab/tree/main/Observation/observation_task_1

8. LEARNING OUTCOME / RESULT:

Successfully implemented dynamic HTML5 Canvas graphics and applied CSS3 layout styling to 'My Styled Webpage'. Demonstrated proficiency in procedural graphics context drawing, modern layout box styling, gradient effects, and interactive CSS hover state transitions.